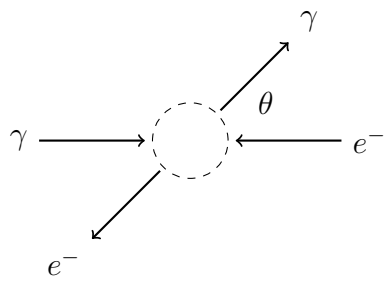
Compton scattering 1

Compton scattering is the process $\gamma + e^- \to \gamma + e^-$.



Momentum vectors where $E = \sqrt{\omega^2 + m^2}$.

$$p_1 = \begin{pmatrix} \omega \\ 0 \\ 0 \\ \omega \end{pmatrix}_{\gamma \longrightarrow} \qquad p_2 = \begin{pmatrix} E \\ 0 \\ 0 \\ -\omega \end{pmatrix}_{\longleftarrow e^-} \qquad p_3 = \begin{pmatrix} \omega \\ \omega \sin\theta \cos\phi \\ \omega \sin\theta \sin\phi \\ \omega \cos\theta \end{pmatrix}_{\nearrow \gamma} \qquad p_4 = \begin{pmatrix} E \\ -\omega \sin\theta \cos\phi \\ -\omega \sin\theta \sin\phi \\ -\omega \cos\theta \end{pmatrix}_{e^- \swarrow}$$

Spinors for p_2.

$$u_{21} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ -\omega \\ 0 \end{pmatrix}_{\text{spin up}} \qquad u_{22} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ 0 \\ \omega \end{pmatrix}_{\text{spin down}}$$

Spinors for p_4.

$$u_{41} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ p_{4z} \\ p_{4x} + ip_{4y} \end{pmatrix}_{\text{spin up}} \qquad u_{42} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ p_{4x} - ip_{4y} \\ -p_{4z} \end{pmatrix}_{\text{spin down}}$$

The scattering amplitude $\mathcal{M}$ is

$$\mathcal{M} = \mathcal{M}_1 + \mathcal{M}_2$$

where

$$\mathcal{M}_1 = \frac{\bar{u}_{4b}(-ie\gamma^\mu)(\not{q}_1 + m)(-ie\gamma^\nu)u_{2a}}{s - m^2}$$

and

$$\mathcal{M}_2 = \frac{\bar{u}_{4b}(-ie\gamma^\nu)(\not{q}_2 + m)(-ie\gamma^\mu)u_{2a}}{u - m^2}$$

Matrices $\not{q}_1$ and $\not{q}_2$ are momentum transfer.

$$\not{q}_1 = (p_1 + p_2)^\mu g_{\mu\nu}\gamma^\nu$$
$$\not{q}_2 = (p_4 - p_1)^\mu g_{\mu\nu}\gamma^\nu$$

Expected density $\langle|\mathcal{M}|^2\rangle$ is the average density.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{ab}|\mathcal{M}|^2 = \frac{1}{4}\sum_{ab}(\mathcal{M}_1+\mathcal{M}_2)(\mathcal{M}_1+\mathcal{M}_2)^*$$

Expand the summand. (Note indices μ and ν of $\mathcal{M}_2$ are interchanged relative to $\mathcal{M}_1$.)

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{ab}\left(\mathcal{M}_1{}^{\mu\nu}\mathcal{M}^*_{1\mu\nu} + \mathcal{M}_1{}^{\mu\nu}\mathcal{M}^*_{2\nu\mu} + \mathcal{M}_2{}^{\nu\mu}\mathcal{M}^*_{1\mu\nu} + \mathcal{M}_2{}^{\nu\mu}\mathcal{M}^*_{2\nu\mu}\right)$$

The summation results are

$$\begin{aligned}
\sum_{ab}\mathcal{M}_1{}^{\mu\nu}\mathcal{M}^*_{1\mu\nu} &= \frac{e^4}{(s-m^2)^2}\left(-8su+24m^2s+8m^2u+8m^4\right)\\
\sum_{ab}\mathcal{M}_1{}^{\mu\nu}\mathcal{M}^*_{2\nu\mu} &= \frac{e^4}{(s-m^2)(u-m^2)}\left(8m^2s+8m^2u+16m^4\right)\\
\sum_{ab}\mathcal{M}_2{}^{\nu\mu}\mathcal{M}^*_{2\nu\mu} &= \frac{e^4}{(u-m^2)^2}\left(-8su+8m^2s+24m^2u+8m^4\right)
\end{aligned}$$

where s, t, and u are Mandelstam variables

$$\begin{aligned}
s &= (p_1+p_2)^2\\
t &= (p_1-p_3)^2\\
u &= (p_1-p_4)^2
\end{aligned}$$

For clarity define

$$\begin{aligned}
f_{11} &= -8su+24m^2s+8m^2u+8m^4\\
f_{12} &= 8m^2s+8m^2u+16m^4\\
f_{22} &= -8su+8m^2s+24m^2u+8m^4
\end{aligned}$$

and write $\langle|\mathcal{M}|^2\rangle$ more compactly as

$$\langle|\mathcal{M}|^2\rangle = \frac{e^4}{4}\left(\frac{f_{11}}{(s-m^2)^2}+\frac{2f_{12}}{(s-m^2)(u-m^2)}+\frac{f_{22}}{(u-m^2)^2}\right)$$

For $E \gg m$ a useful approximation is to set $m=0$ and obtain

$$\begin{aligned}
f_{11} &= -8su\\
f_{12} &= 0\\
f_{22} &= -8su
\end{aligned}$$

Then $\langle|\mathcal{M}|^2\rangle$ simplifies as

$$\langle|\mathcal{M}|^2\rangle = \frac{e^4}{4}\left(-\frac{8su}{s^2}-\frac{8su}{u^2}\right) = -2e^4\left(\frac{u}{s}+\frac{s}{u}\right)$$

For $m = 0$ the Mandelstam variables are

$$\begin{aligned} s &= 4E^2 \\ t &= -2E^2(1 - \cos\theta) \\ u &= -2E^2(1 + \cos\theta) \end{aligned}$$

Hence

$$\langle|\mathcal{M}|^2\rangle = 2e^4 \left(\frac{1 + \cos\theta}{2} + \frac{2}{1 + \cos\theta}\right)$$

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{e^4}\frac{\langle|\mathcal{M}|^2\rangle}{4s} = \frac{\alpha^2}{8E^2}\left(\frac{1 + \cos\theta}{2} + \frac{2}{1 + \cos\theta}\right)$$

Multiply by $(\hbar c)^2$ to convert from natural units to physical units.

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{8E^2}\left(\frac{1 + \cos\theta}{2} + \frac{2}{1 + \cos\theta}\right)(\hbar c)^2$$

Alternate form of $d\sigma$.

$$\frac{d\sigma}{d\cos\theta} = 2\pi\frac{d\sigma}{d\Omega} = \frac{\pi\alpha^2}{s}\left(\frac{\cos\theta + 1}{2} + \frac{2}{\cos\theta + 1}\right)(\hbar c)^2$$

See equation (1) of arxiv.org/pdf/hep-ex/0504012.

Eigenmath script